

Tide retrospective: *generic- (k_1, k_2) 2D separable monomial partition asymptotic*

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Abstract

First sub-tide of v27's generic- (k_1, k_2) 2D programme. Lifts the quartic-sextic 2D partition asymptotic (Z2, 2026-05-18) from $(k_1, k_2) = (2, 3)$ to arbitrary $k_1, k_2 \geq 1$. Four headlines in a new file: exact factorisation through the 1D partitions, exact $\text{const} \cdot t^{-1/(2k_1)-1/(2k_2)}$ form, rescaled `Tendsto`, and `IsEquivalent`. Build clean (8300 jobs); zero `sorry` / `axiom` / `native_decide`.

1 Setting

This session's seventh tide, and the first pivot to the laplace seabed after a long modes-lean arc. The 1D generic-monomial template landed on 2026-05-18 with a parametric closed-form 1D partition function.¹ The 2D `addSeparable` infrastructure provides a factorisation-through-axes lemma.²

The Z2 tide³ packaged this product as an asymptotic at the specific $(k_1, k_2) = (2, 3)$ case. v27's candidate D named the generic- (k_1, k_2) programme as a multi-tide arc; this sub-tide lifts Z2 to parametric k_1, k_2 .

2 The thing formalised

Four theorems in a new `Laplace/TwoD/KthKthPartitionAsymptotic.lean`:

Factorisation.

```
theorem partitionFunction_kthKth_eq
  {k1 k2 : ℕ} (hk1 : 1 ≤ k1) (hk2 : 1 ≤ k2) {t : ℝ} (ht : 0 < t) :
  partitionFunction (addSeparable
    (kthPotential k1) (kthPotential k2)) t =
  partitionFunction (kthPotential k1) t *
  partitionFunction (kthPotential k2) t
```

¹`partitionFunction_kthPotential` in `Laplace/OneD/MonomialPotential.lean`.

²`partitionFunction_addSeparable_factor` in `Laplace/TwoD/AddSeparable.lean`: $Z_{2D}(t) = Z_U(t) \cdot Z_V(t)$ for $L(x, y) = U(x) + V(y)$.

³`Laplace/TwoD/QuarticSexticPartitionAsymptotic.lean`, 2026-05-18.

Exact closed form.

```
theorem partitionFunction_kthKth_eq_const_mul_rpow ... :
  partitionFunction (addSeparable ...) t =
    kthKthPartitionConst k1 k2 *
    t ^ (-(1/(2k1) + 1/(2k2)))
```

Rescaled Tendsto. Multiplied by $t^{1/(2k_1)+1/(2k_2)}$, tends to $C(k_1, k_2)$ at atTop.

IsEquivalent. Asymptotically equivalent to the pure power-law $C(k_1, k_2) \cdot t^{-1/(2k_1)-1/(2k_2)}$.
The composite constant:

$$C(k_1, k_2) := \frac{1}{k_1 k_2} \cdot (2k_1)!^{1/(2k_1)} \cdot (2k_2)!^{1/(2k_2)} \cdot \Gamma(1/(2k_1)) \cdot \Gamma(1/(2k_2)).$$

3 Proof strategy

The structure tracks Z2 exactly with parametric replacement:

1. *Factorisation.* The $n = 0$ specialisation of the 1D polynomial-times-monomial-Gibbs integrability lemma⁴ provides the 1D Boltzmann-weight integrability on each axis. The `addSeparable` factor lemma⁵ then collapses to the product. A small `ext; simp` strips the x^0 on each axis.
2. *Closed form.* Substitute the 1D closed form⁶ on each factor, factor out `Real.div_rpow`, combine the two negative-exponent factors via `Real.rpow_add` and `Real.rpow_neg`, then `ring`.
3. *Tendsto.* The rescaled function is eventually equal to the constant by `add_neg_cancel + Real.rpow_zero`; the `tendsto_const_nhds.congr'` pattern from Z2 carries through.
4. *IsEquivalent.* A `filter_upwards` from the closed form, identical structure to Z2.

4 Roadblocks and resolutions

Nothing. The proof landed on the first build (full Laplace 8300 jobs, zero overfull boxes in the retrospective). The parametric (k_1, k_2) algebra parallels the (2, 3) case exactly once the integrability witnesses are threaded through. The seabed's `push_cast-and-Real.rpow*` tactic vocabulary covered every t -exponent manipulation needed.

5 What was Mathlib, what was new

Mathlib provided the asymptotic API (`Tendsto`, `IsEquivalent`, `filter_upwards`), the `Real.rpow` laws (`div_rpow`, `rpow_add`, `rpow_neg`, `rpow_zero`), and the `tendsto_const_nhds.congr'` pattern. Everything else is laplace-internal.

What was new is four headline theorems plus the composite constant definition, all in a new 149-line file.

⁴`kth_integrable_pow_pot.`

⁵`partitionFunction_addSeparable_factor.`

⁶`partitionFunction_kthPotential.`

6 Lessons

Parametric tides off cache-warm specialisations are cheap. Z2 was the (2, 3) specialisation; this tide is the parametric generalisation. The cache made the mirror-rewrite straightforward; no GPT consult was needed.

First-time pivot to a different seabed in this session takes no incremental setup. The laplace seabed’s documentation (CLAUDE.md with the Mathlib API table and tactic gotcha catalogue) is excellent — a single read sufficed to write the proofs without iteration. The pattern of building per-seabed CLAUDE.md files as living documentation pays back here.

7 Follow-ups

- *Generic- (k_1, k_2) 2D numerator asymptotic.* The Z1 analogue at parametric (k_1, k_2) . Same template (1D moment closed form combined with the partition factorisation). Estimated ~ 150 lines.
- *Generic- (k_1, k_2) 2D moment asymptotic.* The V2 analogue. The 1D `kth_moment_even` times factorisation gives the exact form; asymptotic packagings follow. Estimated ~ 150 lines.
- *Promote the partition factor to **lean-common**.* With a parametric form now landed, the (1D partition $\times$ 1D partition = 2D partition) structural identity is no longer quartic-sextic-specific; deferred until a second-repo consumer.

Acknowledgements

Mirror-shape exemption applied; no GPT-5.5 Pro consult. Z2 (2026-05-18) is the parametric mirror.

Branch `tide/kthkth-partition-asymptotic`, commit `efe5e38` (fast-forwarded into `main`; no separate merge commit).